

Two-Stage Fusion for Routing and Classification in Parameter-Efficient Continual Learning

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Abstract. Task-specific parameter-efficient continual learners must select adapters without knowing the task identity, and better routing need not yield better classification. We augment HRM-PET with a complementary random-projection (RP) head fitted by ridge regression over fixed first-task LoRA features. Its statistics accumulate across tasks without retaining examples or changing the baseline’s adapter training. The same RP scores serve two decision points: task-proposal fusion before re-matching and margin-gated class fusion afterwards. Across three benchmarks and four seeds, the method improves DRM-proposal accuracy by 1.10–2.26 percentage points and classification accuracy by 0.69–1.50. A single-seed grid across four datasets and five pre-training settings shows classification gains in all twenty comparisons. Matched-head controls and conditional ablations distinguish the RP classifier’s contribution from the effects of routing and gating. These effects depend on the dataset and fusion configuration; neither component is uniformly beneficial. The results support improved prediction, not a consistent reduction in forgetting or backward-transfer loss.

Keywords: Continual learning · Parameter-efficient tuning · Parameter routing · Random projection · Signal fusion.

1 Introduction

Continual learning asks a model to absorb a stream of tasks without losing what earlier ones taught it. Parameter-efficient tuning adapts a frozen pre-trained backbone using a small number of parameters. One important family maintains task-specific prompts or adapters and selects them at inference [5, 3]. Such methods can avoid storing old examples, but preserving parameters does not guarantee preserved accuracy.

In this family, task identity is unknown at inference and parameter selection is itself a prediction. A wrong adapter can produce unsuitable features. Larger pools, attention-weighted mixtures and explicit re-matching offer different responses to this difficulty [4, 7, 1]. We focus on HRM-PET, where routing and classification use related predictions from the adapted and unadapted backbone.

We add a complementary predictor: first-task adapter features, a frozen random projection and a ridge classifier fitted from accumulated statistics, following

RanPAC [2]. Fitting uses current-task data without changing LoRA/TII training or retaining examples. Its scores revise the task proposal before re-matching and blend with final class logits, with greater weight when the primary classifier’s top-two margin narrows.

Contributions. Our contribution is the integration and evaluation of a shared analytic predictor at two decision points, not a new ridge-learning algorithm. First, we introduce task-proposal fusion before re-matching and margin-gated class fusion afterwards, reusing one RP head. Second, matched-feature controls and four-seed conditional ablations test whether routing and gating add value beyond class fusion alone. Third, a four-dataset, five-pre-training grid and weight-sensitivity study examine the scope of the gains. We distinguish classification improvements from retention effects and oracle complementarity from achieved corrections.

2 Related Work

Parameter pools and the selection they require. L2P [4] and Dual-Prompt [5] use queries from a frozen backbone to select lightweight parameters from a pool. CODA-Prompt [7] combines prompts using attention weights. HiDe-Prompt [3] separates within-task prediction, task-identity inference and task-adaptive prediction, identifying identity inference as a bottleneck. HRM-PET [1] uses LoRA adapters [6] and re-matching to revisit initial selections. MoTE [14] combines task-specific experts; DLEPEM [15] combines general and adapted prototypes for matching.

Closed-form solutions in continual learning. Closed-form classifiers preserve past contributions through additive statistics of fixed features. APER [8] adapts the backbone on the first task, then uses frozen features and class prototypes. ACIL [20] uses recursive least squares, replacing a sample buffer with correlation statistics and recovering the corresponding joint solution. DS-AL [21] adds a second analytic stream. RanPAC [2] applies a fixed random projection to frozen features, accumulates second-order statistics and solves a ridge system.

We inherit RanPAC’s projection, Gram accumulation and ridge solve, but retain the adapter-routing pipeline and consult the head at two decision points instead of using it as the sole predictor.

Correcting the classifier after the fact. Class-incremental learning often biases predictions towards recent classes [9]. iCaRL [10] uses nearest exemplar means; BiC [11] learns an affine logit correction on held-out data; Weight Aligning [12] matches new- and old-class weight norms; and LUCIR [13] normalises features and weights. Our fusion leaves the adapters, auxiliary classifier and shared head fixed. It retains no exemplars but requires current-task data to fit the RP head.

2.1 HRM-PET: The Baseline We Build On

We use the released HRM-PET [1] inference path. Its two re-matching stages provide the decision points that our method augments.

Task-identity inference (TII). The *bare* backbone ϕ , without an adapter, produces $q(x) = \phi(x; \theta_{\text{ptm}})$. A cross-entropy-trained auxiliary classifier gives logits $u = g(q(x); \theta_\omega)$ and the initial task match $t_0 = \mathcal{T}(\arg \max_c u_c)$.

Re-matching (DRM and CRM). The first adapted pass attaches p_{t_0} and produces $r = g_{\theta_g}(\phi(x; p_{t_0}, \theta_{\text{ptm}}))$. DRM proposes the task of $\arg \max r$ and evaluates its adapter when it differs from t_0 ; our fusion modifies this proposal, not the preceding bare TII pass. CRM selects low-confidence samples from r by the baseline’s generalised-entropy threshold and compares the current logits, under the baseline log-sum-exp score, with the adapter of the second-ranked class of r ; it is confidence-triggered, not restricted to cases where the two task identities agree. Both reuse trained modules, can require further adapter forwards, and are heuristics: neither guarantees that accuracy cannot decrease.

Shared classification head. Each adapted pass uses the same classification head over all seen classes. Let $\ell \in \mathbb{R}^{|\mathcal{Y}|}$ denote the logits retained after re-matching, predicting $\hat{y} = \arg \max_c \ell_c$; unseen classes are masked with $-\infty$. CRM may replace these logits without updating the task identity recorded by the evaluator (Sect. 4.1). The baseline therefore leaves two opportunities for complementary evidence: revising the DRM proposal and revising the final class prediction.

3 Method

3.1 Problem Statement

Class-incremental learning (CIL) aims to learn T tasks while mitigating forgetting. Task t provides labelled images $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ from class set $\mathcal{C}_t$, with $\mathcal{C}_t \cap \mathcal{C}_{t'} = \emptyset$ for $t \neq t'$. Only current-task data are available for training; past images are never retained. At inference task identity is unknown and prediction spans all seen classes.

Our task-specific PET setting [3, 1] freezes backbone parameters θ_{ptm} and maintains LoRA adapters $\mathcal{P} = \{p_1, \dots, p_T\}$, one per task, with shared classification head g_{θ_g} . The primary prediction requires adapter selection; our fixed RP branch is independent of the inferred task.

We distinguish four class-score vectors: bare TII scores u , initial adapted shared-head logits r , final re-matched logits ℓ , and RP scores s . The class-to-task map is $\mathcal{T}(\cdot)$; t_0 is the initial TII match and $\hat{t}_{\text{DRM}}$ the task proposed for direct re-matching. We write $z(\cdot)$ for per-sample standardisation with the population standard deviation floored at 10^{-6} .

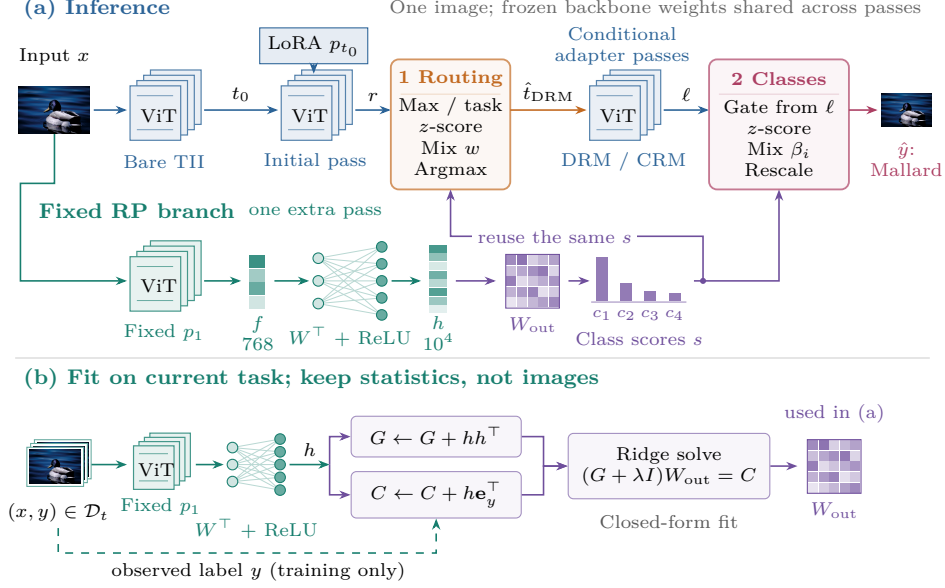


Fig. 1. Two-stage fusion. All inference passes use the same image; adapted passes share head g . TII selects t_0 ; the projection W is frozen. Stage 1 standardises task-wise maxima of r and s ; Stage 2 uses the top-two probability margin of ℓ for its continuous gate (Eqs. 6–8), then predicts by argmax. DRM runs if $\hat{t}_{\text{DRM}} \neq t_0$; CRM uses low-confidence r and its second-ranked class’s task, comparing logits by log-sum-exp (not at task 1). Only seen classes/tasks participate; score bars are schematic. Photo: T. G. Barnes/USFWS (public domain).

3.2 The Second Source of Evidence

We fit an RP head after each task and reuse its scores in two inference-time fusion operations. HRM-PET parameters are not retrained. Figure 1 distinguishes the initial adapted logits from the bare TII scores and shows both reuse paths.

The RP head follows RanPAC [2]. Features $f \in \mathbb{R}^d$, $d = 768$, are column vectors from ViT-B/16 with the first task’s LoRA adapter (p_1 here, index 0 in code), fixed during accumulation and inference and unnormalised; changing adapters between tasks would mix incompatible spaces.

Projection and second-order statistics. The feature is projected into a high-dimensional space of size $M = 10^4$ and passed through a nonlinearity, and the projected features are accumulated into two matrices:

$$h = \text{ReLU}(W^\top f) \in \mathbb{R}^M, \quad W \in \mathbb{R}^{d \times M}, \quad W_{jk} \sim \mathcal{N}(0, 1), \quad (1)$$

$$G = \sum_n h_n h_n^\top \in \mathbb{R}^{M \times M}, \quad C = \sum_n h_n \mathbf{e}_{y_n}^\top \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad (2)$$

where $\mathbf{e}_y$ is the basis vector of class y . Each current-task training sample contributes once; G and C are accumulated in float64. W is generated once with seed 1993 and then frozen.

The nonlinear expansion permits richer decision functions than a linear head on f . Without ReLU, $W^\top f$ remains linear in f and adds no separating power. This does not guarantee that the expanded data become linearly separable.

Classification by ridge regression. The output weights are the closed-form solution of a regularised least-squares problem:

$$W_{\text{out}} = (G + \lambda I)^{-1} C \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad s = W_{\text{out}}^\top h \in \mathbb{R}^{|\mathcal{Y}|}. \quad (3)$$

The implementation solves the regularised system rather than inverting it.

Two inherited properties. For a fixed feature map, the sums in (2) preserve earlier contributions and yield the joint ridge solution over all observed features, independent of sample order in exact arithmetic. Storage is independent of sample count and retains no per-sample features. This does not imply bitwise floating-point invariance or zero forgetting: new classes can outscore old ones. Changing the first task also changes its adapter and thus the feature map.

3.3 Two-Stage Fusion

Both stages blend sources with different score scales. Direct mixing can make the nominal weights poorly reflect their influence. Each source is standardised per sample:

$$z(v) = \frac{v - \text{mean}(v)}{\sigma_\epsilon(v)}, \quad \sigma_\epsilon(v) = \max\{\text{std}(v), 10^{-6}\}, \quad (4)$$

where std is the population standard deviation. Moments are over seen tasks in Stage 1 and seen classes in Stage 2, excluding unseen entries.

First stage: fusion at the routing level. Stage 1 runs after the initial adapted pass and before DRM. Its input is r , the shared-head logits under the adapter selected by bare TII, not auxiliary logits u . Both r and RP scores s are mapped to seen tasks by $r_t^{\text{task}} = \max_{c \in \mathcal{C}_t} r_c$ and $s_t^{\text{task}} = \max_{c \in \mathcal{C}_t} s_c$ for $t = 1, \dots, \hat{T}$, with $\hat{T}$ the number of tasks seen so far; classes of unseen tasks never enter either maximum. The two vectors are blended after standardisation:

$$\text{fused} = w z(r^{\text{task}}) + (1 - w) z(s^{\text{task}}), \quad \hat{t}_{\text{DRM}} = \arg \max_t \text{fused}_t, \quad w = 0.7. \quad (5)$$

For one seen task the standard deviation is zero, so the floor keeps the operation finite. The fused task replaces the baseline DRM proposal $\mathcal{T}(\arg \max r)$; the initial TII match, CRM candidate rule, confidence test and shared head are unchanged. At $w = 1$ the ordering is that of r^{task} , recovering that proposal; the full baseline also needs $\beta = 0$.

Second stage: gated fusion at the classification level. Because ℓ comes from the shared head, it can predict the correct class in spite of a wrong routing decision, so we use the projection head itself as a secondary classifier, this time taking its class scores s . A fixed blend would alter confident primary predictions; we instead reduce the RP share as the top-two probability margin grows, while recognising that confident predictions can still be wrong:

$$\beta_i = \beta \left(1 - \frac{p_{(1)} - p_{(2)}}{p_{(1)} + p_{(2)}} \right), \quad \beta = 0.5, \quad (6)$$

where $p_{(1)} \geq p_{(2)}$ are the two largest probabilities of $\text{softmax}(\ell)$ over the classes seen so far. Dividing by the sum makes the quantity depend only on the ratio $p_{(2)}/p_{(1)}$, so the gate measures how threatened the top class is rather than how much mass the pair holds. The RP share falls to zero as the top class dominates and reaches β when the two are equal: a continuous weight, not a binary test.

The blend reuses (4) and then maps the result back onto the scale of ℓ :

$$m_i = (1 - \beta_i) z(\ell_i) + \beta_i z(s_i), \quad (7)$$

$$\hat{\ell}_i = m_i \cdot \sigma_\epsilon(\ell_i) + \text{mean}(\ell_i) \in \mathbb{R}^{|\mathcal{Y}|}. \quad (8)$$

The positive affine map-back preserves rankings and restores a reference scale for cross-entropy, not a calibration guarantee: for standardised nondegenerate vectors with correlation ρ and the floor inactive, the mixture variance is $(1 - \beta_i)^2 + \beta_i^2 + 2\beta_i(1 - \beta_i)\rho$, which need not equal one. At $\beta = 0$ this stage leaves ℓ unchanged.

Cost. At inference the head costs one extra adapter forward per sample, serving both stages. That is the head’s own cost, not the pipeline’s difference, which we did not measure: changing the DRM proposal can change the number of adapter evaluations.

Per task the head accumulates G and C over the current task’s training split and solves one ridge system. Its stored arrays scale as $O(M^2 + M|\mathcal{Y}| + dM)$: G has M^2 entries, C and W_{out} each have $M|\mathcal{Y}|$, and the projection has dM . Storage does not grow with the number of observed samples for fixed dimensions and class capacity. The Gram matrix dominates in our setting. At $M = 10^4$ it holds 10^8 float64 entries, or 800 MB. Together with its regularised copy and Cholesky factor, these three matrices require approximately 2.4 GB; this is an analytical estimate, not a measured total-memory peak. A historical ten-task evaluation on ImageNet-R took 283 s on one RTX 4090 at batch 24; neither its phase-level costs nor its overhead over the baseline were measured.

4 Experiments

4.1 Setup

The backbone is ViT-B/16 [22], pre-trained on ImageNet-21K. Rank-8 LoRA modifies the key/value projections in the first five blocks. The four-seed study

splits ImageNet-R [23], CIFAR-100 [24] and CUB-200-2011 [25] into 10 tasks of 20, 10 and 20 classes, respectively. The grid replaces CUB with ImageNet-A [26] (10×20 classes) and the five-dataset protocol of [1] (one domain per task). Its pre-training settings are Sup-21K, MoCo-v3 [27], iBOT-1K/iBOT-21K [28] and DINO [29].

Each seed (42–45) has separately trained LoRA/TII parameters; pre-trained backbone weights remain frozen. Both methods use identical saved checkpoints and evaluation paths. RP fits current-task training images once using the evaluation transform: resize to 256, centre crop to 224, and convert to a tensor; CIFAR additionally uses its configured channel normalisation. No test labels enter the fit. We use $M = 10^4$, ReLU, $\lambda = 10^4$, no additional RP feature normalisation or temperature calibration, and no fusion ramp. Fusion starts at task 1; CRM retains **En=gen**, threshold -10 , and the second-ranked class candidate. Training/evaluation scripts and dataset preprocessing are versioned in the code repository [30].

Acc@task in the released evaluator measures the DRM task proposal. In the default class-based routing path, CRM can change the output logits without updating the task identity passed to this metric; it is therefore not a measure of final adapter selection after CRM. Acc@1, Acc@5 and Loss use the final classification logits. Final metrics average the per-task values with equal task weights, not by pooling all test images. Writing $a_{i,t}$ for classification accuracy on task i at stage t , the retention measures are

$$\text{Forgetting} = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{i \leq t \leq T} a_{i,t} - a_{i,T} \right), \quad (9)$$

$$\text{Backward} = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{i,T} - a_{i,i}). \quad (10)$$

Both averages exclude the last task, and the maximum includes only stages at which task i has been learned. Lower is better for Loss and Forgetting.

Reproducibility and sanity check. $w = 1$ restores the baseline DRM proposal and $\beta = 0$ bypasses class fusion. With corrected unseen-class masking, this identity matches the baseline to logged precision at every stage for all three datasets and seeds 42–45. For example, ImageNet-R seed 42 gives Acc@task 77.7914, Acc@1 74.0191 and Loss 1.2305 in both paths. This checks that bypassing fusion restores the evaluated baseline, not that intermediate fusion operations are harmless. Re-evaluation confirmed the reported final accuracies of the main runs, controls, grids, ablations and diagnostics. Intermediate losses and retention changed in some earlier logs, including retention in ten full-method grid runs; corrected values are used throughout. All forty grid evaluations were checked; backbone/first-adapter feature tensors match across checkpoints in all main/grid checkpoint sets.

4.2 Main Results

Table 1 reports all three datasets over four seeds. The change column is pairwise — taken per seed, then averaged — so its standard deviation describes the stability of the improvement, not the seeds’ difficulty.

Table 1. Final results, mean $\pm$ sample SD over seeds 42–45; Change is paired full minus baseline. Accuracy/retention are percentages (changes in points).

Dataset	Metric	HRM-PET	Proposed	Change
<i>ImageNet-R</i>	Acc@task	77.58 ± 0.33	79.85 ± 0.18	$+2.264 \pm 0.158$
	Acc@1	73.94 ± 0.48	75.18 ± 0.24	$+1.236 \pm 0.416$
	Acc@5	86.71 ± 0.21	88.02 ± 0.17	$+1.312 \pm 0.148$
	Loss	1.23 ± 0.01	1.19 ± 0.01	-0.040 ± 0.002
	Forgetting	3.36 ± 0.56	3.30 ± 0.35	-0.056 ± 0.334
	Backward	-3.16 ± 0.61	-3.20 ± 0.42	-0.047 ± 0.282
<i>CIFAR-100</i>	Acc@task	89.77 ± 0.11	90.87 ± 0.09	$+1.103 \pm 0.066$
	Acc@1	89.78 ± 0.03	90.47 ± 0.05	$+0.690 \pm 0.063$
	Acc@5	98.65 ± 0.03	98.86 ± 0.07	$+0.218 \pm 0.057$
	Loss	0.39 ± 0.00	0.39 ± 0.00	-0.0003 ± 0.0012
	Forgetting	3.89 ± 0.08	3.74 ± 0.12	-0.150 ± 0.095
	Backward	-3.87 ± 0.07	-3.74 ± 0.11	$+0.136 \pm 0.095$
<i>CUB-200</i>	Acc@task	93.12 ± 0.14	94.34 ± 0.10	$+1.226 \pm 0.149$
	Acc@1	86.38 ± 0.25	87.88 ± 0.27	$+1.495 \pm 0.353$
	Acc@5	97.04 ± 0.10	97.44 ± 0.13	$+0.400 \pm 0.103$
	Loss	0.57 ± 0.00	0.53 ± 0.01	-0.036 ± 0.004
	Forgetting	2.54 ± 0.35	2.18 ± 0.12	-0.359 ± 0.325
	Backward	-2.42 ± 0.39	-2.14 ± 0.13	$+0.279 \pm 0.391$

Intervals are paired 95% Student- t , $\bar{\Delta} \pm 3.182 s_{\Delta} / \sqrt{4}$: imprecise at four seeds, and assuming approximately normal seed-wise differences. Correlated metrics mean we discuss axes rather than count confirmations.

Routing (Acc@task): improved. The gain is +1.10 to +2.26 points, with positive paired intervals on all three datasets. This is the axis the first stage acts on directly.

Classification (Acc@1, Acc@5): improved. Acc@1 rises by +0.69 to +1.50 points and Acc@5 by +0.22 to +1.31, with positive paired intervals for both metrics on all three datasets. The full method’s Acc@1 improvement repeats across every seed. These aggregate gains do not establish how many routing repairs also repair classification.

Retention (Forgetting, Backward): unresolved. All six paired retention intervals include zero. CIFAR Forgetting is $-0.150 [-0.302, +0.002]$ and Back-

ward $+0.136 [-0.015, +0.287]$. These related metrics are not independent confirmations. Neither consistent improvement nor equivalence is established: four seeds leave substantial uncertainty, and no equivalence margin was specified. The single-seed decomposition in Sect. 4.4 is descriptive.

4.3 Placement Against Prior Methods

Table 2 places the method among the baselines the original work reports. Its first eight rows are quoted from Table 1 of [1] under Sup-21K, averages over three seeds; the last two are our own runs at seed 42 through the code path of Sect. 4.1.

Table 2. Final average accuracy (%) under Sup-21K. Rows above the rule are quoted from Table 1 of [1], averaged over three seeds; the two below are ours at seed 42, from a different implementation and seed set. Bold indicates the best value within each group, not a controlled cross-group ranking. S-Prompt++ and HiDe-LoRA are the variants evaluated in [1].

Method	CIFAR-100	ImageNet-R	ImageNet-A	5-Datasets
DualPrompt [5]	86.60	68.79	39.76	76.09
S-Prompt++ [16]	88.81	69.68	39.10	83.38
CODA-Prompt [7]	86.94	70.03	30.96	62.54
LAE [17]	85.10	72.25	40.36	64.16
CPrompt [18]	83.18	69.24	37.60	71.72
InfLoRA [19]	85.28	72.24	42.12	71.97
HiDe-LoRA [3]	88.43	71.90	42.18	93.25
HRM-PET [1]	89.45	73.86	44.28	93.99
HRM-PET, our run	89.77	74.02	45.85	91.37
Proposed	90.53	75.51	47.42	93.77

Our seed-42 baseline differs from the published mean by $+0.32$, $+0.16$, $+1.57$ and -2.62 points on the four datasets, respectively. These reproduction gaps preclude a controlled comparison across the two groups. The controlled comparison remains the paired baseline–full difference in Table 1.

4.4 Generalisation Across Pre-training Settings

Table 3 reports the 4×5 grid using one code path at seed 42. All twenty cells improve Acc@1 by $+0.59$ to $+3.58$ points, against baseline Acc@1 of 31.8–93.8; Acc@task gains are $+0.97$ to $+4.57$.

Final accuracy on earlier tasks rises in every cell ($+1.83$ on average), versus $+1.58$ at learning time. Backward, their difference, worsens in five cells: learning-time accuracy rose more, not final accuracy less. This decomposition uses rounded per-task logs and remains descriptive.

Table 3. Acc@1 change (percentage points), full minus matched HRM-PET, seed 42. All cells use the same $w = 0.7$, $\beta = 0.5$ and margin gate; these are single-seed comparisons, not confidence intervals.

Pre-training	ImageNet-R	CIFAR-100	ImageNet-A	5-Datasets
Sup-21K	+1.49	+0.76	+1.57	+2.40
MoCo-v3	+1.89	+2.12	+2.41	+1.41
iBOT-1K	+2.52	+2.06	+3.58	+0.85
iBOT-21K	+2.55	+1.80	+2.63	+0.59
DINO	+2.68	+2.03	+0.97	+0.75

4.5 Ablation

Table 4 isolates the matched-feature RP predictor and the class-fusion variants. All RP variants use the same fixed first-task features and ridge configuration. RP-only predicts directly from s and is not a reproduction of the complete RanPAC protocol. Class-only disables routing fusion ($w = 1$); both class-only variants use $\beta = 0.5$.

Table 4. Component comparison: final Acc@1 (%), mean $\pm$ sample SD over seeds 42–45. Full uses $w = 0.7$, $\beta = 0.5$ and the margin gate. The full method is not uniformly best among the variants.

Variant	ImageNet-R	CIFAR-100	CUB-200
HRM-PET	73.94 $\pm$ 0.48	89.78 $\pm$ 0.03	86.38 $\pm$ 0.25
Same-feature RP-only	69.89 $\pm$ 0.17	88.80 $\pm$ 0.20	87.26 $\pm$ 0.27
Class-only, ungated	74.49 $\pm$ 0.14	90.15 $\pm$ 0.16	88.01 $\pm$ 0.21
Class-only, gated	75.10 $\pm$ 0.21	90.50 $\pm$ 0.08	87.72 $\pm$ 0.26
Full	75.18 $\pm$ 0.24	90.47 $\pm$ 0.05	87.88 $\pm$ 0.27

Full exceeds RP-only on all three datasets, but class fusion accounts for much of the gain over HRM-PET. Ungated class-only has the highest CUB mean and gated class-only the highest CIFAR mean; we retain one configuration rather than select a different winner on each dataset.

Routing with ungated class fusion. The five-backbone ImageNet-R ablation uses seed 42; baseline Acc@1 ranges from 68.47 to 74.02. We report differences between specified variants. Routing-only and ungated class-only variants improve Acc@1 on each backbone by +0.29–+0.73 and +0.51–+3.07, respectively. Adding routing to ungated class fusion changes Acc@1 by -0.07 to $+0.02$, not establishing equivalence.

Full-minus-gated-class-only Acc@1 is +0.13, +0.29, +0.78, +0.43 and +0.77 on Sup-21K, MoCo-v3, iBOT-1K, iBOT-21K and DINO at seed 42. These single-seed gains establish neither reliability nor retention benefit: MoCo-v3 Forgetting

risers from 3.44 to 3.69. The gate is not uniformly beneficial either: full minus ungated full is -0.26 Acc@1 on MoCo-v3 and -0.54 on iBOT-1K.

Conditional contributions over four seeds. Full minus gated class-only measures conditional routing; gated minus ungated class-only measures the gate without routing fusion. These are not independent stage effects.

The conditional routing effect is $+0.083$ $[+0.026, +0.140]$ on ImageNet-R, -0.030 $[-0.085, +0.025]$ on CIFAR-100 and $+0.156$ $[+0.055, +0.257]$ on CUB-200. The gate is $+0.603$ and $+0.355$ on the first two, both excluding zero, and -0.284 $[-0.570, +0.002]$ on CUB-200. Each pair sums to the full-against-ungated-class-only change, $+0.69$, $+0.33$ and -0.13 , to within 0.005. Routing is positive on two datasets and unresolved on CIFAR. The gate’s CUB estimate is negative but its interval includes zero; this does not establish harm. Full still improves over HRM-PET in Table 1. Only the first fusion stage changes Acc@task.

4.6 Complementarity Diagnostics

These seed-42 diagnostics use equal task weights (Sect. 4.1). On ImageNet-R, bare TII routing accuracy is 63.80%, the baseline DRM proposal 77.79% and the RP router 77.12%. The TII–RP oracle union reaches 82.99%. Dividing the RP-correct/TII-wrong rate by TII’s error rate gives 53.0%, versus 62.0% on CIFAR-100 and 46.9% on CUB-200. These are task-balanced recovery ratios, not pooled image counts or achieved fusion accuracy. They compare bare TII with RP, not Stage 1’s adapted logits with RP.

Complementarity also exists at the class level. In the routing-only diagnostic ($w = 0.7$, $\beta = 0$), ImageNet-R accuracy is 74.31% for the routed head and 70.08% for RP, with an oracle union of 78.89%. RP is correct when the routed head is wrong on 4.58 task-balanced percentage points, versus 2.43 on CIFAR-100 and 3.87 on CUB-200. This potential motivates reusing s for classification, but does not count actual repairs or harms of full fusion.

4.7 Sensitivity to the Mixing Weights

The two weights were fixed sequentially, w before the second stage existed and β afterwards, so the second stage may have shifted the optimum of the first. Table 5 reports Acc@1 over the joint grid on ImageNet-R at seed 42 with the margin gate active.

The grid favours $\beta = 0.5$ at every w ; its maximum is $w = 0.6$, not the reported 0.7. At $\beta = 0.5$, Acc@1 spans 0.11 points, versus about one point for Acc@task. Across four seeds, 0.6 gains $+0.332$ $[+0.143, +0.522]$ Acc@task and $+0.108$ $[-0.064, +0.280]$ Acc@1 over 0.7 with the gate and $\beta = 0.5$; the Acc@1 sign reverses on seed 43. These full-method comparisons establish neither equivalence nor routing-only optimality. We retain the earlier $w = 0.7$ for consistency, not as a known optimum.

Table 5. Acc@1 over the $w \times \beta$ grid, ImageNet-R at seed 42, full method with the margin gate.

	$\beta = 0.3$	$\beta = 0.5$	$\beta = 0.7$
$w = 0.6$	75.28	75.61	75.29
$w = 0.7$ (reported)	75.22	75.51	75.24
$w = 0.8$	75.12	75.50	75.27

4.8 Limitations

ImageNet-R and CUB informed development; neither is an untouched holdout. More seeds do not remove selection bias. CUB/pre-training overlap [25] was not audited. The grids and five-backbone ablation are single-seed; only the three-dataset contrasts use four seeds. CIFAR’s conditional routing effect is unresolved, and gate benefits vary across backbones. Same-protocol RanPAC reproduction remains absent despite the matched-head control in Table 4. Quoted baselines use another implementation; our 5-Datasets result is below the published figure. Retention benefit/equivalence remains unestablished; phase-level cost and actual full-fusion repair/harm counts were not measured. The loader skips two zero-byte NotMNIST training files, also empty in the local source archive; we did not replace them.

5 Conclusion

Reusing one fixed-feature RP predictor for task proposals and class scores improves HRM-PET by 0.69–1.50 Acc@1 points across three datasets and four seeds, without changing adapter training or storing past images. Component comparisons show that class fusion provides much of the gain; the added value of routing and gating is conditional, not universal. The method improves prediction but does not establish reduced forgetting. Controlled cost measurements and independent validation are priorities for further evaluation.

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